shrink the number of virtual machines, you need to decouple your EC2 instances from the clients, because the interface that's reachable from outside the system needs to stay the same no matter how many EC2 instances are working behind the scenes.

Figure 17.5 shows how to build a scalable system based on synchronous or asynchronous decoupling. A load balancer acts as the entry point for synchronous decoupling, by distributing requests among a fleet of virtual machines. A message queue is used as the entry point for asynchronous requests, and messages from producers are stored in the queue. The virtual machines then poll the queue and process the messages asynchronously.

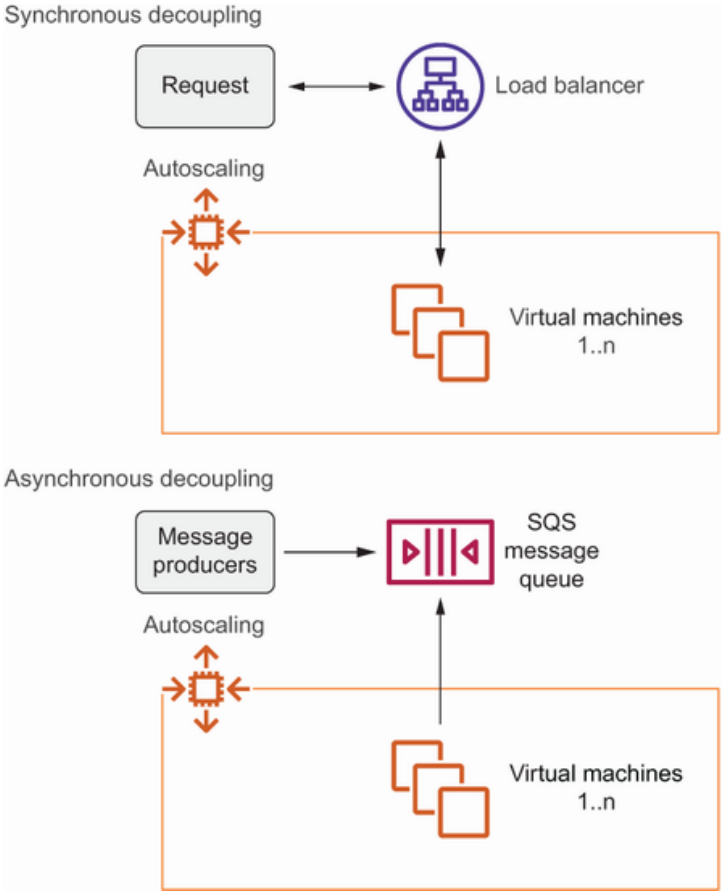


Figure 17.5 Decoupling allows you to scale the number of virtual machines dynamically.

Decoupled and scalable applications require stateless servers. A stateless server stores any shared data remotely in a database or storage system. The following two examples implement the concept of a stateless server: